

Stochastic Dynamic Processes

Module 2 - Part I: Foundations

Synthesis of Mathematical Foundations for Poisson and Markov Processes.

1. Stochastic Processes: Definition

A **stochastic process** is a family of random variables $(X_t)_{t \in T}$.

- **Index Set T :** Represents time.
 - *Discrete Time:* $T = \mathbb{N}$ (e.g., daily stock prices).
 - *Continuous Time:* $T = \mathbb{R}_+$ (e.g., signal noise).
- **State Space:** The set of all possible values X_t can take.
- **Realization:** A single observed path $x(t)$ of the process.

2. Probability Axioms (Kolmogorov)

For a sample space S and an event A :

1. **Non-negativity:** $P(A) \geq 0$.
2. **Normalization:** $P(S) = 1$.
3. **Additivity:** For disjoint events A, B , $P(A \cup B) = P(A) + P(B)$.

Combinatorics Refresh

- **Permutations** (Order matters): $P_n^r = \frac{n!}{(n-r)!}$
- **Combinations** (Order doesn't matter): $C_n^r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

3. Conditional Probability & Bayes

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ (where } P(B) > 0 \text{)}$$

Bayes' Theorem

Updating beliefs based on new evidence B :

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

4. Discrete Random Variables

Distribution	PMF: $P(X = k)$	$E(X)$	$Var(X)$
Bernoulli(p)	$p^k(1 - p)^{1-k}$	p	$p(1 - p)$
Binomial(n, p)	$\binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Poisson(λ)	$\frac{e^{-\lambda} \lambda^k}{k!}$	λ	λ
Geometric(p)	$(1 - p)^{k-1} p$	$1/p$	$(1 - p)/p^2$

5. Continuous Random Variables

Exponential Distribution (λ)

Critical for modeling waiting times.

- **PDF:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- **Memoryless Property:** $P(X > s + t | X > t) = P(X > s)$.
 - *The remaining wait time is independent of how long you have already waited.*

Normal Distribution (μ, σ^2)

- Defined by the Gaussian bell curve.
- Central Limit Theorem: The sum of independent R.V.s tends toward Normality.

6. Stationarity

A process is **Stationary** if its statistical properties do not change over time.

Weak-Sense Stationarity (WSS)

1. **Constant Mean:** $E[X(t)] = \mu$ (constant).
2. **Autocorrelation:** $E[X(t_1)X(t_2)]$ depends only on the lag $\tau = t_1 - t_2$.

Strict-Sense Stationarity

The entire joint distribution is invariant to time shifts.

7. Poisson Process Properties

1. **Stationary & Independent Increments:** Events in disjoint intervals are independent.
2. **Aggregation:** Sum of two independent Poisson processes (λ_1, λ_2) is a Poisson process with rate $\lambda_1 + \lambda_2$.
3. **Disaggregation:** If events are filtered with probability p , the resulting process is Poisson with rate λp .
4. **Inter-arrival Times:** For a Poisson process with rate λ , the times between events are **Exponentially distributed** with parameter λ .